Representations of Lie Groups
A Mathematical Dissection of Quantum Spin
Part I: Mathematical Foundations

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Outline

Topological Spaces

Definition (topological space)

A **topological space** is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T} \subseteq \mathscr{P}(X)$ satisfies:

1. (Trivial bounds) $\varnothing \in \mathcal{T}$ and $X \in \mathcal{T}$.
2. (Arbitrary unions) $\forall \{U_\alpha\}_{\alpha \in \mathcal{I}} \subseteq \mathcal{T} : \bigcup_{\alpha \in \mathcal{I}} U_\alpha \in \mathcal{T}$.
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Key constructions. Product topology, subspace topology, metric topologies (e.g. the Euclidean topology on $\mathbb{R}^n$).

Topological Manifolds

Definition (topological manifold)

A **topological manifold** is a topological space $(M, \mathcal{T})$ satisfying:

1. (Hausdorff) Points can be separated by disjoint open sets.
2. (Second-countable) The topology has a countable basis.
3. (Locally Euclidean) Every point has a neighbourhood homeomorphic to $\mathbb{R}^n$ for some fixed $n = \dim(M)$.

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Why Hausdorff + second-countable? These are technical conditions that prevent pathological spaces. They guarantee that partitions of unity exist.

Smooth Manifolds

Definition (smooth manifold)

A **smooth manifold** is a triple $(M, \mathcal{T}, \mathcal{A})$ where:

1. $(M, \mathcal{T})$ is a topological manifold.
2. $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \mathcal{I}}$ is a **smooth atlas**: each $(U_\alpha, \varphi_\alpha)$ is a chart, the charts cover M, and transition maps $\varphi_\beta \circ \varphi_\alpha^{-1}$ are C^∞.
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Key idea. Charts are local coordinate systems. Smoothness means that when you switch between coordinate systems, the change-of-coordinates map is infinitely differentiable.

Smooth Maps and Diffeomorphisms

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A map $F : M \to N$ between smooth manifolds is **smooth** if its coordinate representation $\psi \circ F \circ \varphi^{-1}$ is C^∞ for all compatible charts.

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Example. $\mathbb{R}^n$ is diffeomorphic to the open unit ball $B_1(0)$. S^n is *not* diffeomorphic to $\mathbb{R}^n$ (compact vs. non-compact).

Tangent Spaces via Derivations

Motivation

How do we define tangent vectors to a smooth manifold without embedding it in $\mathbb{R}^N$? Answer: **derivations** — linear maps that satisfy the Leibniz rule.

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Definition (tangent space)

Let M be a smooth manifold, $p \in M$. The **stalk** $C_p^\infty M$ is the set of germs of smooth functions at p. The **tangent space** at p is

$$T_pM \overset{\text{df}}{=} \left\{ v : C_p^\infty M \to \mathbb{R} \;:\middle|:\; v \text{ is a derivation at } p \right\}.$$

T_pM is an $\mathbb{R}$-vector space with $\dim(T_pM) = \dim(M)$.

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Why derivations? For $\mathbb{R}^n$, a tangent vector v acts on functions by the directional derivative: $v(f) = \sum_i v^i \frac{\partial f}{\partial x^i}\Big|_p$. This algebraic characterisation generalises to any manifold.

The Differential

Definition (differential)

Let $F : M \to N$ be smooth, $p \in M$. The **differential of F at p** is the linear map $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ defined by

$$\forall v \in T_pM,\ \forall [f]_{F(p)} \in C^\infty_{F(p)}N : \quad [(\mathrm{d}F_p(v))]\Big([f]_{F(p)}\Big) \stackrel{\text{df}}{=} v\Big([f \circ F]_p\Big).$$

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Interpretation. The differential is the "best linear approximation" to a smooth map, generalising the Jacobian matrix from multivariable calculus.

Groups, Fields, and Vector Spaces

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A set G with an associative binary operation μ, an identity e, and inverses ι.
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Why these matter. Lie groups combine the smooth structure of manifolds with the algebraic structure of groups. Lie algebras are vector spaces with an extra bracket operation.

Algebras and Lie Algebras

Definition ($\mathbb{F}$-algebra)

A vector space $\mathcal{A}$ over $\mathbb{F}$ together with a bilinear multiplication $\mathcal{A} \times \mathcal{A} \to \mathcal{A}$. (Examples: $\mathbb{R}^{n,n}$ with matrix multiplication, $C^\infty(M)$ with pointwise multiplication.)

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An $\mathbb{F}$-vector space $\mathfrak{g}$ with a bilinear bracket $[\bullet, \bullet] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ satisfying:

1. (Skew-symmetry) $[Y, X] = -[X, Y]$
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Key example. $(\mathbb{R}^3, \times)$ — the cross product makes $\mathbb{R}^3$ a Lie algebra. $[v, w] = v \times w$ satisfies skew-symmetry and Jacobi.

Lie Groups

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A **Lie group** is a smooth manifold $(G, \mathcal{T}, \mathcal{A})$ that is also a group $(G, \{\mu, \iota, e\})$ such that:

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Examples.

- $(\mathbb{R}^n, +)$ — the translation group.
- $\mathrm{GL}(n, \mathbb{R})$ — invertible $n \times n$ real matrices.
- $\mathrm{SO}(n)$ — rotations in $\mathbb{R}^n$.
- $\mathrm{SU}(n)$ — unitary transformations with determinant 1.

Left Translation and Left-Invariant Vector Fields

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Geometric meaning. A left-invariant vector field is completely determined by its value at the identity e. Translating the vector at e by L_g gives the vector at any point g.

The Lie Algebra of a Lie Group

Theorem (tangent space at identity is a Lie algebra)

Let G be a Lie group. Define $\mathfrak{g} = T_e G$ (tangent space at identity). For each $v \in \mathfrak{g}$, there exists a **unique** left-invariant vector field X^v with $X^v\big|_e = v$. Define the bracket on $\mathfrak{g}$ by

$$[v, w]_{\mathfrak{g}} \stackrel{\text{df}}{=} [X^v, X^w]_{\circ}\big|_e.$$

Then $\left(\mathfrak{g}, [\bullet, \bullet]_{\mathfrak{g}}\right)$ is a Lie $\mathbb{R}$-algebra with $\dim(\mathfrak{g}) = \dim(G)$.

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Remark. We denote the Lie algebra of a Lie group G by $\mathrm{Lie}(G)$.

The Closed Subgroup Theorem (É. Cartan)

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Let G be a Lie group and $H \subseteq G$ a closed subgroup. Then H admits a unique smooth structure making it a Lie group such that the inclusion $H \hookrightarrow G$ is a smooth embedding.

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Example. $\mathrm{SO}(n)$, $\mathrm{SU}(n)$, $\mathrm{SL}(n, \mathbb{R})$, and $\mathrm{Sp}(2n)$ are all closed subgroups of $\mathrm{GL}(N, \mathbb{F})$ and therefore Lie groups.

The General Linear Group $\mathrm{GL}(n, \mathbb{F})$

Example (the general linear group)

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- Its Lie algebra is $\mathfrak{gl}(n, \mathbb{F}) = \mathbb{F}^{n,n}$ (all $n \times n$ matrices) with the commutator bracket:

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- Most matrix Lie groups arise as closed subgroups of $\mathrm{GL}(n, \mathbb{F})$. Their Lie algebras are subspaces of $\mathfrak{gl}(n, \mathbb{F})$ closed under the commutator.

The Special Orthogonal Group SO(3)

Example (the special orthogonal group — rigid rotations)

$$\mathrm{SO}(3) = \left\{ \boldsymbol{R} \in \mathbb{R}^{3,3} \;:\!|\!:\; \boldsymbol{R}^\intercal \boldsymbol{R} = \boldsymbol{I}_3,\ \det(\boldsymbol{R}) = 1 \right\};$$

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$\dim(\mathfrak{so}(3)) = \dim(\mathrm{SO}(3)) = 3$, with basis $[\boldsymbol{L}_i, \boldsymbol{L}_j] = \varepsilon_{ijk}\boldsymbol{L}_k$:

$$\boldsymbol{L}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix},\ \boldsymbol{L}_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix},\ \boldsymbol{L}_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

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Remark. $[\boldsymbol{L}_i, \boldsymbol{L}_j] = \varepsilon_{ijk}\boldsymbol{L}_k$ — the same structure as the cross product on $\mathbb{R}^3$.

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Remark. $[\boldsymbol{T}_i, \boldsymbol{T}_j] = \varepsilon_{ijk}\boldsymbol{T}_k$ — **identical** to $\mathfrak{so}(3)$. This is the first hint of the isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

The Double Cover $\Phi : \mathrm{SU}(2) \to \mathrm{SO}(3)$

Construction

For $\boldsymbol{U} \in \mathrm{SU}(2)$, define $\Phi(\boldsymbol{U}) : \mathfrak{su}(2) \to \mathfrak{su}(2)$ by

$$[\Phi(\boldsymbol{U})](\boldsymbol{X}) \stackrel{\mathrm{df}}{=} \boldsymbol{U}\boldsymbol{X}\boldsymbol{U}^\dagger.$$

Using the basis $\beta = (\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{T}_3)$, identify $\mathfrak{su}(2) \cong \mathbb{R}^3$. Then the matrix $[\Phi(\boldsymbol{U})]_\beta^\beta$ lies in $\mathrm{SO}(3)$.

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For $\boldsymbol{U} \in \mathrm{SU}(2)$, define $\Phi(\boldsymbol{U}) : \mathfrak{su}(2) \to \mathfrak{su}(2)$ by

$$[\Phi(\boldsymbol{U})](\boldsymbol{X}) \stackrel{\text{df}}{=} \boldsymbol{U}\boldsymbol{X}\boldsymbol{U}^\dagger.$$

Using the basis $\beta = (\boldsymbol{T}_1, \boldsymbol{T}_2, \boldsymbol{T}_3)$, identify $\mathfrak{su}(2) \cong \mathbb{R}^3$. Then the matrix $[\Phi(\boldsymbol{U})]_\beta^\beta$ lies in $\mathrm{SO}(3)$.

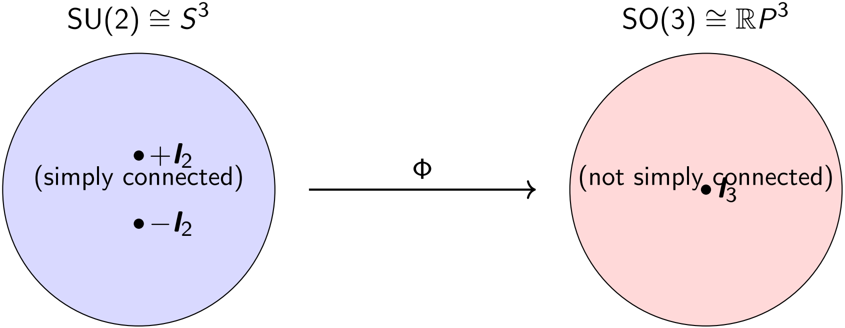
Theorem (double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$)

$\Phi : \mathrm{SU}(2) \to \mathrm{SO}(3)$ is:

1. A smooth group homomorphism: $\Phi(\boldsymbol{U}\boldsymbol{V}) = \Phi(\boldsymbol{U}) \circ \Phi(\boldsymbol{V})$.
2. Surjective.
3. Has kernel $\ker(\Phi) = \{\pm \boldsymbol{I}_2\}$ (two elements).

Thus $\mathrm{SU}(2)$ is a **double cover** of $\mathrm{SO}(3)$.

The Double Cover: Geometric Picture



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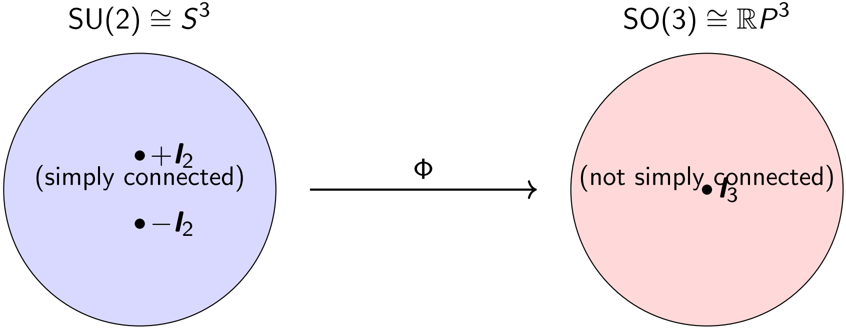

Why two-to-one? Both $\boldsymbol{U}$ and $-\boldsymbol{U}$ give the same rotation. A rotation by angle θ about axis $\hat{\boldsymbol{n}}$ lifts to $\pm\cos(\theta/2)\boldsymbol{I}_2 \mp \mathrm{i}\sin(\theta/2)(\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma})$. Increasing θ by 2π flips the sign.

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Theorem (Lie algebra isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$)

Define $\varphi : \mathfrak{so}(3) \to \mathfrak{su}(2)$ by $\varphi(\boldsymbol{L}_i) \stackrel{\text{df}}{=} \boldsymbol{T}_i$ (and extend $\mathbb{R}$-linearly). Then φ is a Lie algebra isomorphism.

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Proof (essence)

- φ maps a basis to a basis $\implies$ linear isomorphism.
- Both satisfy $[\boldsymbol{L}_i, \boldsymbol{L}_j] = \varepsilon_{ijk} \boldsymbol{L}_k$ and $[\boldsymbol{T}_i, \boldsymbol{T}_j] = \varepsilon_{ijk} \boldsymbol{T}_k$.
- Therefore $\varphi([\boldsymbol{L}_i, \boldsymbol{L}_j]) = \varphi(\varepsilon_{ijk} \boldsymbol{L}_k) = \varepsilon_{ijk} \boldsymbol{T}_k = [\boldsymbol{T}_i, \boldsymbol{T}_j] = [\varphi(\boldsymbol{L}_i), \varphi(\boldsymbol{L}_j)]$.

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Significance. Although SU(2) and SO(3) are different as groups (double cover vs. quotient), their Lie algebras are **identical**. Infinitesimal rotations are the same in both pictures.

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6. SO(3) (rotations) and SU(2) (unitary 2×2 matrices) are the central examples.
7. SU(2) is a **double cover** of SO(3). Their Lie algebras are **isomorphic**: $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

What Comes Next (Part II)

Representation theory

- Representations of Lie algebras and Lie groups.
- Complexification $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$ and the Cartan–Weyl basis $(\boldsymbol{H}, \boldsymbol{E}, \boldsymbol{F})$.
- Classification of irreducible representations of $\mathfrak{su}(2)$: for each $j \in \left\{0, \frac{1}{2}, 1, \frac{3}{2}, \ldots\right\}$, a unique irrep of dimension $2j + 1$.
- Integration: every $\mathfrak{su}(2)$-representation exponentiates to an SU(2)-representation.
- Projective representations and the descent to SO(3): integer $j \to$ ordinary representations, half-integer $j \to$ spinor (projective) representations.
- Physical spin operators and the 4π periodicity of spinors.